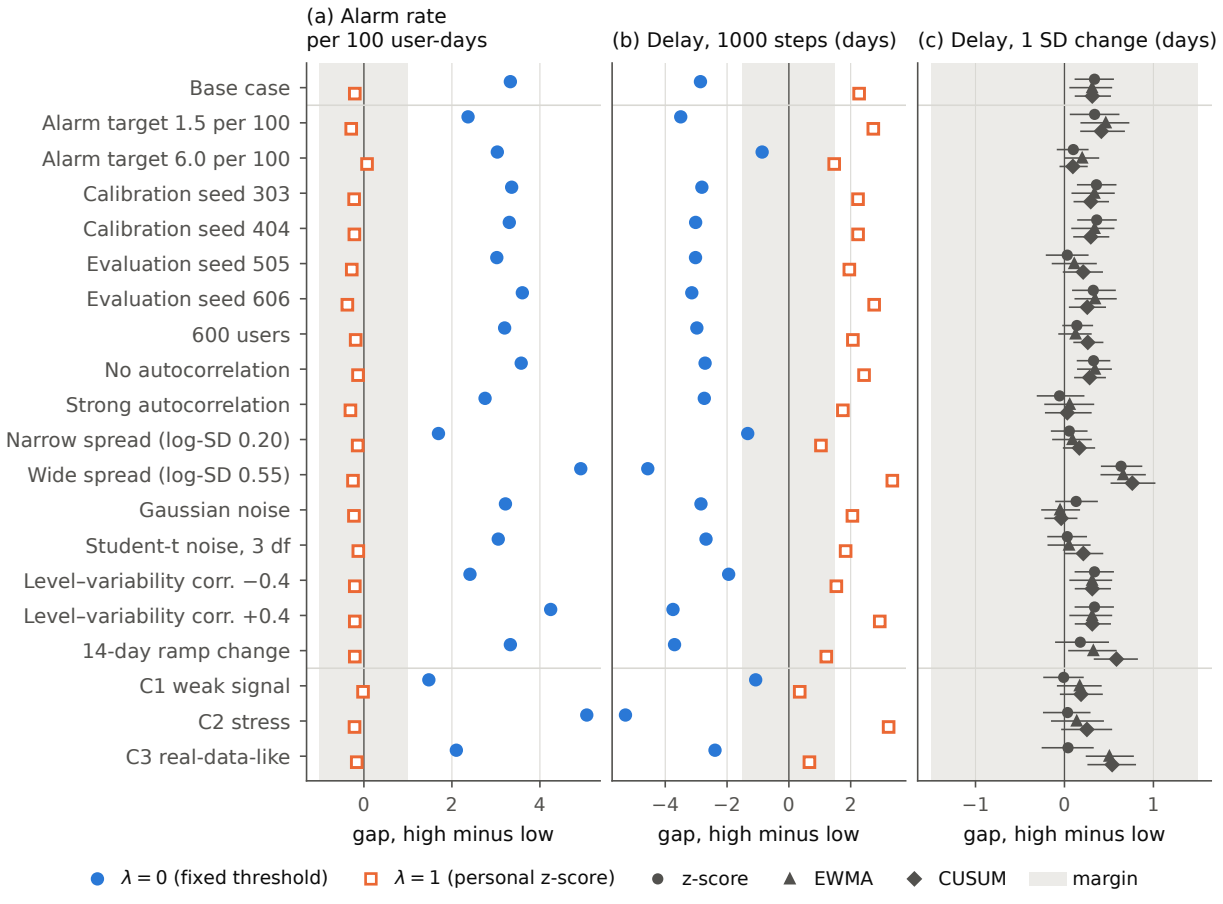
(a) Alarm rate per 100 user-days
(b) Delay, 1000 steps (days)
(c) Delay, 1 SD change (days)
Base case
Alarm target 1.5 per 100
Alarm target 6.0 per 100
Calibration seed 303
Calibration seed 404
Evaluation seed 505
Evaluation seed 606
600 users
No autocorrelation
Strong autocorrelation
Narrow spread (log-SD 0.20)
Wide spread (log-SD 0.55)
Gaussian noise
Student-t noise, 3 df
Level–variability corr. −0.4
Level–variability corr. +0.4
14-day ramp change
C1 weak signal
C2 stress
C3 real-data-like
0
2
4
−4
−2
0
2
−1
0
1
gap, high minus low
gap, high minus low
gap, high minus low
λ = 0 (fixed threshold)
λ = 1 (personal z-score)
z-score
EWMA
CUSUM
margin